EXPERIMENTS IN MODERN PHYSICS

Second Edition

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ACADEMIC PRESS

An imprint of Elsevier Science

Amsterdam Boston London New York Oxford Paris San Diego
San Francisco Singapore Sydney Tokyo

1.4. THE HYDROGEN SPECTRUM

The hydrogen atom is the simplest quantum-mechanical system. It consists of an electron bound, due to the Coulomb force, to a proton. It is characteristic of bound quantum-mechanical systems that their total energy cannot have any value, but that the system is found in one of a discrete set of energy levels, or states. Transitions of the system between these states may occur. Such transitions must satisfy the basic conservation laws of electric charge, energy, momentum, angular momentum, and the other relevant symmetries of nature.

Transition from a higher energy state to a state with less energy can occur for an isolated system, and the larger the probability for this transition, the shorter the "lifetime" of that excited state. During such spontaneous transitions of a quantum-mechanical system to a lower energy state, a quantum of radiation, or one or more particles, can be emitted, which will carry away the energy lost by the system (after recoil effects have been taken into account). In the presence of a radiation field the quantum-mechanical system can either gain energy from the field and change into a state with higher energy, or lose energy to the field and revert to a lower energy state. For all quantum-mechanical systems there exists a lowest energy state, called the *ground state*.

By observing the quanta of radiation, or the particles emitted during such transitions, we gain information on the energy levels involved. The typical example is optical spectroscopy, which consists of the accurate determination of the energy of the light quanta emitted by atoms. Infrared spectroscopy deals mainly with the quanta emitted by molecules, nuclear spectroscopy with the quanta emitted in nuclear transitions, and so on. In nuclei, however, the separation between energy levels is much larger, so that the emitted quanta of electromagnetic radiation lie in the gamma ray region; thus different techniques are employed for detection and measurement of their energy. It is also very common for nuclei to decay from one energy state to another by the emission of an electron and neutrino (beta decay) and for certain heavier nuclei by the emission of a helium nucleus (alpha particle). Similar processes take place in the interactions or decay of the elementary particles.

The idea of energy levels and their structure for the hydrogen atom was first introduced by Niels Bohr in 1913. However, a complete theoretical interpretation had to wait until the introduction of the Schrödinger equation in 1926. Even then, for theory to agree with observation it is necessary to

include additional small effects such as the fine and hyperfine structure, relativistic motion, and other higher order corrections. These corrections are derived using the theory of quantum electrodynamics (QED) so that today we can theoretically calculate the energy levels of the hydrogen atom to the amazing accuracy of 1 part in 10^{11}.

We will use the Bohr theory to predict the hydrogen energy levels, because it is so simple, even though it assigns the incorrect angular momentum to the states. The postulates of the Bohr theory are (a) that the electron is bound in a circular orbit around the nucleus such that the angular momentum is quantized in integral units of Planck's constant (divided by 2π); namely, $pr = mvr = n(h/2\pi) = n\hbar$; and (b) that the electron in this orbit does not radiate energy, unless a transition to a different orbit occurs. We can then calculate the radii of these orbits and the total energy of the system, potential plus kinetic energy of the electron. The attractive force between the electron (charge $-e$) and the proton (charge $+e$) or a nucleus (of charge $+Ze$) is the Coulomb force, which is set equal to the centripetal force.

The total mechanical energy of the electron is

$$\begin{aligned} E &= T + V \\ &= \frac{1}{2}mv^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}. \end{aligned} \tag{1.7}$$

Here m, v, and $-e$ are the electron's mass, velocity, and electric charge, $+Ze$ is the charge on the nucleus, and r is the "orbital radius" of the electron.[10] The potential energy, of course, is just the attractive Coulomb potential between the electron and the nucleus. We can relate the velocity v to the other variables by using $F = ma$, where F is the Coulomb force and a is the centripetal acceleration. That is

$$\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} = m\frac{v^2}{r},$$

which implies that

$$v^2 = \frac{1}{m}\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}. \tag{1.8}$$

[10] We assume that the nucleus is infinitely heavy.

If we introduce this result into Eq. (1.7) we obtain

$$E = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} = -\frac{1}{2}\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} = -\frac{1}{2}|V|. \tag{1.9}$$

At this point we can impose the Bohr quantization condition

$$r = n\frac{\hbar}{mv} \tag{1.10}$$

to eliminate v in Eq. (1.8). Here n is the principal *quantum number*. We obtain

$$\frac{n^2\hbar^2}{m^2r^2} = \frac{1}{m}\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$$

or

$$\frac{1}{r} = \frac{m}{n^2\hbar^2}\frac{1}{4\pi\varepsilon_0}Ze^2. \tag{1.11}$$

Inserting this result in Eq. (1.9) we find for the total energy

$$E_n = -\left[\frac{mZ^2e^4}{2(4\pi\varepsilon_0)^2\hbar^2}\right]\frac{1}{n^2}. \tag{1.12}$$

For the hydrogen atom where $Z = 1$, the expression in brackets in Eq. (1.12) equals 13.6 eV. This is the energy required to take an electron in the ground state ($n = 1$) and separate it from the nucleus completely ($E = 0$). We refer to it as the binding energy of the hydrogen atom. It is customary to introduce the Rydberg constant (wave number) through

$$E_n = -hcR_\infty\frac{1}{n^2}, \tag{1.13}$$

where

$$R_\infty = 10973731.534\ \mathrm{m}^{-1}$$

and thus

$$E_1 = -13.6057\ \mathrm{eV}.$$

Furthermore, from Eq. (1.11) we can write for the radius of the orbits in hydrogen

$$r_n = n^2a_\infty$$

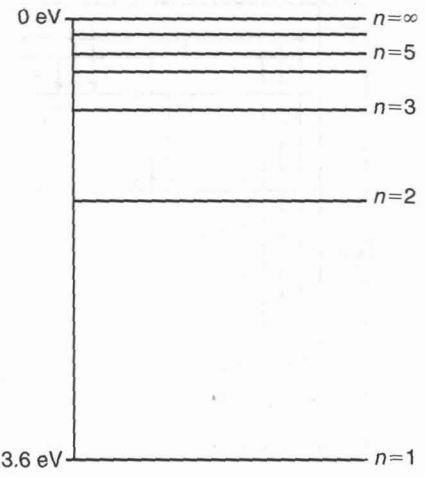


FIGURE 1.12 Energy-level diagram of the hydrogen atom according to the simple Bohr theory.

with

$$a_\infty = \frac{\hbar^2}{m}\frac{4\pi\epsilon_0}{e^2} = 0.5291772 \times 10^{-10}\ \text{m},$$

called the Bohr radius.

The energy levels of the hydrogen atom that we derived can be represented by Fig. 1.12. However, the lines observed in the spectrum correspond to transitions between these levels; this is shown in Fig. 1.13, where arrows have been drawn for all possible transitions. The energy of a line is given by

$$\Delta E_{if} = hcR_\infty\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), \tag{1.14}$$

where the subscripts i and f stand for initial and final state, respectively.

Since the frequency of the radiation is connected to the energy of each quantum through

$$E = h\nu$$

one finds that

$$\frac{1}{\lambda} = \frac{\nu}{c} = \frac{E}{hc}$$

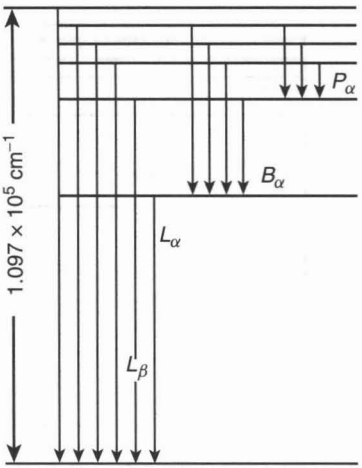


FIGURE 1.13 Transitions between the energy levels of a hydrogen atom. The lines L_α, L_β, etc., belong to the Lyman series, B_α, B_β, etc., to the Balmer series, and P_α, P_β, etc., to the Paschen series, and so forth.

and

$$\frac{1}{\lambda_{if}} = R_\infty \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right). \tag{1.15}$$

Indeed, the simple expression of Eq. (1.15) is verified by experiment to a high degree of accuracy.

From Eq. (1.14) (or from Fig. 1.13) we note that the spectral lines of hydrogen will form groups depending on the final state of the transition, and that within these groups many common regularities will exist; for example, in the notation of Fig. 1.13

$$\nu(L_\beta) - \nu(L_\alpha) = \nu(B_\alpha).$$

If $n_f = 1$, then

$$\lambda_{i1} = 91.1 \left(\frac{n_i^2}{n_i^2 - 1} \right) \text{nm} \qquad n_i \geq 2$$

and all lines fall in the far ultraviolet; they form the (so-called) Lyman series. Correspondingly if $n_f = 2$, then

$$\lambda_{i2} = 364.4 \left(\frac{n_i^2}{n_i^2 - 4} \right) \text{nm} \qquad n_i \geq 3$$

and all lines fall in the visible part of the spectrum, forming the Balmer series. For $n_f = 3$ the series is named after Paschen and falls in the infrared.

1.5. EXPERIMENT ON THE HYDROGEN SPECTRUM

1.5.1. General

To measure the frequency of the radiation emitted by atoms one can use either a grating or a prism to disperse the different wavelengths. When using a prism, one exploits the variation, with wavelength, of the refractive index of certain media. Prism spectrometers are limited to wavelength regions for which they are able to transmit the radiation; for example, in the infrared, special fluoride or sodium chloride prisms and lenses are used. In the ultraviolet, the optical elements are made of quartz. Also, the sensitivity of the detectors varies with wavelength, so that different types are used in each case (thermopile, photographic emulsion, phototube, etc.).

In this laboratory a small constant-deviation prism spectrograph and a 2-in. reflection grating spectrometer were used. We will consider in detail a measurement of the hydrogen spectrum with the grating, since an absolute value for the wavelengths can be obtained and visual detection is used. A brief discussion of prism spectrographs is given in Section 1.5.4.

From Fig. 1.14, it is evident that the path difference between rays 1 and 2 after reflection is

$$BD - AC = CB \sin \theta_r - CB \sin \theta_i,$$

where CB is the grating spacing d. The angles θ_i and θ_r are both taken as positive when they lie on opposite sides of the normal. Since for constructive interference the path difference must be a multiple of the wavelength, we obtain the condition

$$n\lambda = d\ (\sin \theta_r - \sin \theta_i). \tag{1.16}$$

It can be shown[11] that the resolution of the grating is given by

$$\frac{\lambda}{\Delta\lambda} = nN,$$

where n is the order of diffraction and N the total number of rulings. The same considerations apply to a transmission grating.

[11] See Chapter 5, Section 5.5.

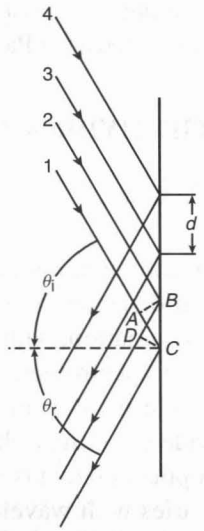


FIGURE 1.14 Schematic diagram of a reflection grating. A parallel beam of radiation is incident along the rays 1 through 4 at an angle θ_i, with respect to the normal; the reflected radiation is observed at an angle θ_r. The spacing between the grooves of the grating is d.

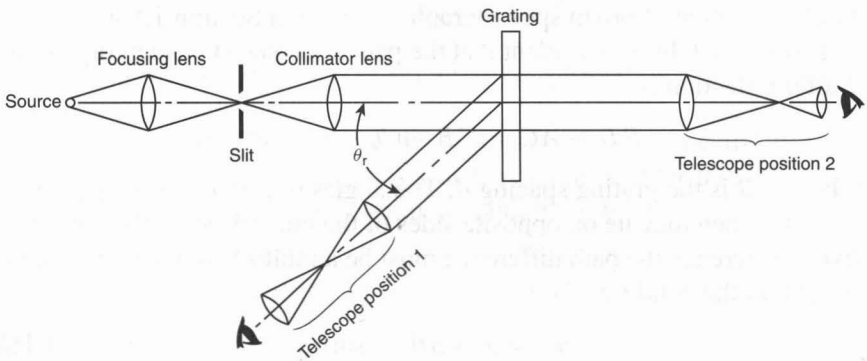


FIGURE 1.15 Diagrammatic arrangement of a grating spectrometer.

The grating is mounted on a goniometer table in the general arrangement shown in Fig. 1.15. A slit and collimating lens are used to form a beam of parallel light from the source, and a telescope mounted on a rotating arm is used for viewing the diffracted lines. It is obviously necessary to ensure

parallelism of the incident and reflected beams, normality of the grating, and so on. A suggested alignment procedure is as follows:

(a) The viewing telescope is focused for parallel rays (on some distant object).

(b) Then with the grating removed, the slit is viewed with the telescope (in position 2) to ascertain that the slit is aligned and in focus; in this way the collimator lens is adjusted.

(c) The source and source lens are placed in position and the alignment and focusing are again checked. The cross hairs are aligned with the slit.

(d) This position of the telescope is carefully noted since it represents the 0° position. The readings on the scale should be made to one minute of a degree by using the vernier and a flashlight.

(e) From now on one may have to work in dark, or by draping the apparatus with a black cloth.

(f) The grating is placed in position and aligned for normal incidence ($\theta_i = 0$). This can be done by "autocollimation"; a strong light is focused onto the slit and a cardboard mask with a narrow slit is placed on the collimator lens. The grating is then adjusted until the reflected image of the cardboard slit coincides with the slit itself.

(g) Finally, the lines of the grating should be made parallel to the slit (hence the cross hairs); this can be done by viewing one edge of the grating with the telescope in position 1.

With any reasonable grating it is possible to observe the visible lines of the spectrum in several orders; thus we expect the measurements for λ/d to be self-consistent, since

$$\sin\theta_{m+1} - \sin\theta_m = (m+1)\frac{\lambda}{d} - m\frac{\lambda}{d} = \frac{\lambda}{d} \tag{1.17}$$

independently of angle of incidence θ_i, or order.[12] The grating spacing d is usually stated by the manufacturer; for example, the grating in this laboratory had rulings on the order of 7000 to the inch ($d = 3.629 \times 10^{-6}$ m). However, d can be obtained by using one or more standard lines of known wavelength.

[12]Provided that both θ_m and θ_{m+1} are taken on the same side of the normal.

The following data were obtained by a student using the grating spectrometer. The source was a low-pressure hydrogen discharge tube (Cenco type 87210) operated at a few thousand volts; a 5-kV transformer and variac were used to provide the variable voltage. The useful life of these discharge tubes is limited because of the appearance of strong molecular bands after some hours of operation.

1.5.2. Determination of d

To obtain the grating spacing d, sodium (Na) was used as a standard, and measurement on three lines (for the shorter wavelength of the doublet) gave the results shown in Table 1.2. Since for all the above measurements θ_i is the same, it follows that

$$d^{-1}(n_k\lambda_k) + \sin\theta_i = \sin\theta_k$$

and a least-squares fit to the linear relation $\beta x + \alpha = y$ can be made; we have

$$\frac{1}{d} = \frac{N\sum(n_k\lambda_k \sin\theta_k) - \sum(\sin\theta_k)\sum(n_k\lambda_k)}{N\sum(n_k\lambda_k)^2 - \left[\sum(n_k\lambda_k)\right]^2}, \tag{1.18}$$

TABLE 1.2 Diffraction Angles from a Sodium Source

λ in nm	Order n	θ_n	$\theta_i = 19°12'$
615.43	1	29°42′	
	2	41°27′	
	3	55°58′	
589.00	1	29°14′	
	2	40°21′	
	3	53°49′	
	4	75°15′	
568.27	2	39°32′	
	3	52°12′	
	4	70°48′	

where the sums are over k, $k = 1, 2, \ldots, N$ and N is the total number of measurements. From the data of Table 1.2 we obtain[13]

$$\frac{1}{d} = 2.7085 \pm 0.009 \times 10^5 \text{ m}^{-1} \tag{1.19}$$

in good agreement with the manufacturer's specification.

Some care must be exercised when comparing wavelengths, since they do depend on the refractive index, n, of the medium in which they are measured,

$$c' = \frac{c(\text{vacuum})}{n},$$

hence

$$\lambda' = \frac{\lambda(\text{vacuum})}{n}.$$

The wavelengths listed in most tables are given for dry air at a pressure of 760 mm mercury. However, any theoretical calculation, such as in Eq. (1.15) predicts the vacuum wavelengths. The refractive index of air at stp is

$$n(\text{air}) = 1.00029. \tag{1.20}$$

1.5.3. The Balmer Series

Measurements on the first four members of the Balmer series, which lie in the visible region, can be made with the spectrometer described above. The data obtained by a student and their reduction are given in Table 1.3.

We observe that the obtained values for the wavelengths of the Balmer series are in agreement with the accepted values at the level of 1/1000. We can now test Eq. (1.15) and obtain the Rydberg wave number. We note that

$$\frac{1}{\lambda} = R_{\text{H}}\left[\frac{1}{4} - \frac{1}{n^2}\right].$$

So that from a least-squares fit

$$R_{\text{H}} = \frac{\sum \rho_i^2}{\sum \lambda_i \rho_i},$$

[13] In reaching this result we have constrained $\theta_{\text{i}} = 19°12'$.

TABLE 1.3 Data on the Balmer Series of Hydrogen as Obtained with a Grating Spectrometer

Color	θ_n	$\sin\theta_n - \sin\theta_i$	Order	Calculated λ	Accepted λ	Balmer series identification
Violet	33°12′	0.22199	2	410.75 ± 6	410.17 H_δ	$n_i = 6$
	41°15′	0.33378	3			
Blue	26°16′	0.11698	1			
	34°06′	0.23483	2	433.82 ± 8	434.05 H_γ	$n_i = 5$
	42°42′	0.35259	3			
Green	27°10′	0.13001	1			
	36°04′	0.26316	2	485.75 ± 10	486.13 H_β	$n_i = 4$
	46°09′	0.39559	3			
Red	30°11′	0.17720	1			
	42°57′	0.35579	2	657.94 ± 14	656.28 H_α	$n_i = 3$
	59°29′	0.53532	3			

Note. All wavelengths are in nm. These measurements used $d = 3692.1 \pm 30$ nm as determined by the previous measurements on the sodium standard lines, and $\sin\theta_i = 0.32557$.

where

$$\rho_i = \frac{4n_i^2}{n_i^2 - 4},$$

giving

$$R_H = (1.09601 \pm 0.003) \times 10^7 \text{ m}^{-1}$$

in good agreement with the accepted value[14]

$$R_H = \frac{M}{M+m} R_\infty = 1.096776 \times 10^7 \text{ m}^{-1}.$$

Here M is the mass of the proton and m the mass of the electron.

1.5.4. The Prism Spectrograph

Long before gratings became widely available, prisms were used as the dispersive element in spectrographs. Prism spectrographs are handy for viewing a large span of the spectrum and come in various ingenious optical

[14]The difference between R_H and R_∞ is due to the motion of the electron about the center of mass rather than about the proton.

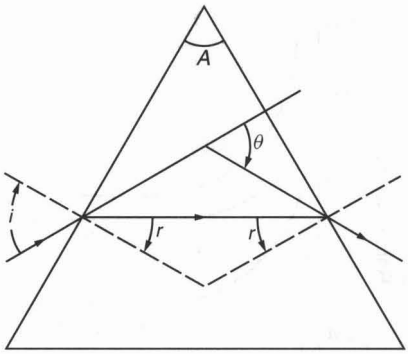


FIGURE 1.16 Diffraction of a ray at minimum deviation through a prism of apex angle A.

arrangements. The dispersion of a prism is a function of the refractive index; thus it cannot be used for absolute measurements without careful calibration.

In the case of a simple prism at minimum deviation (see Fig. 1.16) the diffraction angle θ is given by

$$\frac{\sin\theta_i}{\sin\theta_r} = n \qquad 2\theta_r = A \qquad \theta_i = \theta_r + \frac{1}{2}\theta;$$

thus

$$\sin\left(\frac{A+\theta}{2}\right) = n\sin\frac{A}{2}, \tag{1.21}$$

where θ_i and θ_r are the angles of incidence and refraction, respectively, and A is the apex of the prism. In Fig. 1.17 the refractive index of flint glass as a function of wavelength is given. We note that in the determination of wavelength from the diffraction angle the relation is by no means linear and is in general of serious complexity. Further, most modern prism spectrographs do not consist of a single dispersive element, but of some combination of prisms. The instrument used in this laboratory was of the "constant-deviation" type, and Fig. 1.18 gives the optical paths for an incident ray. It may be seen that the angle of incidence and the angle of exit can remain fixed for all wavelengths by an appropriate rotation of the prism; this has obvious advantages for positioning and alignment of source and detector.

The rotation of the prism is calibrated to give rough wavelength indications, but measurements are made on the exposed photographic plate

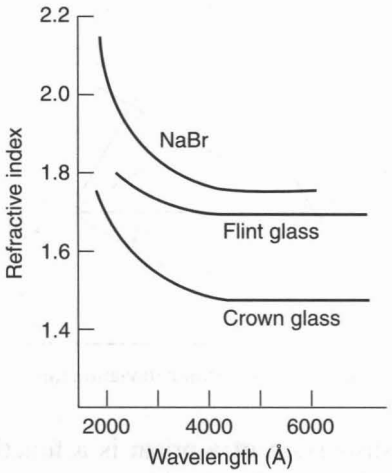


FIGURE 1.17 Refractive index of various materials as a function of wavelength.

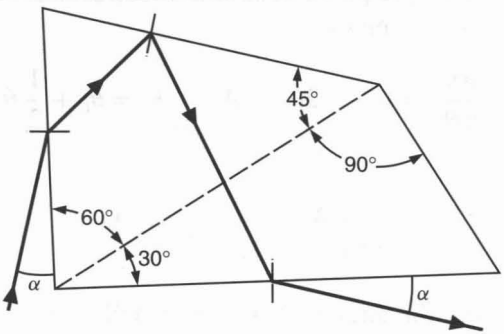


FIGURE 1.18 A constant-deviation prism and the diffraction of a ray passing through it.

or film. A known spectrum is superimposed on the spectrum that is to be investigated, and an interpolation between the known lines is used.

The general arrangement of the spectrograph is shown in Fig. 1.19. Source, lens, and slit should be aligned and the source focused on the slit. By viewing through the eyepiece and varying the prism position, one can get a feeling for the dispersion and the range of the instrument. To obtain photographs of a spectrum, the telescope is replaced by the camera assembly. Several exposures can be had on the same plate; to distinguish different spectra superimposed at the same location on the plate, the "fishtail," which controls the length of the slit, can be used.

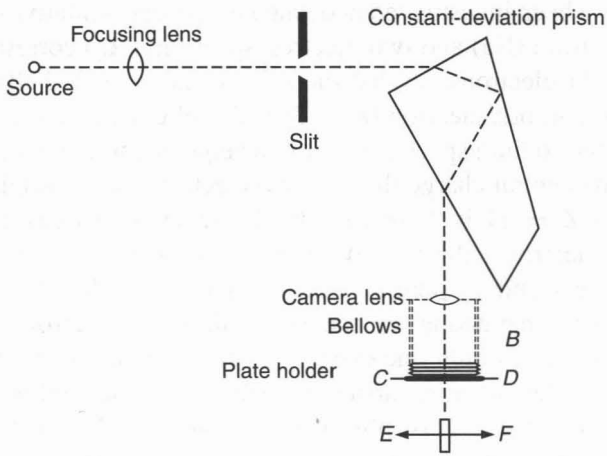


FIGURE 1.19 Schematic arrangement of the constant-deviation spectrograph.

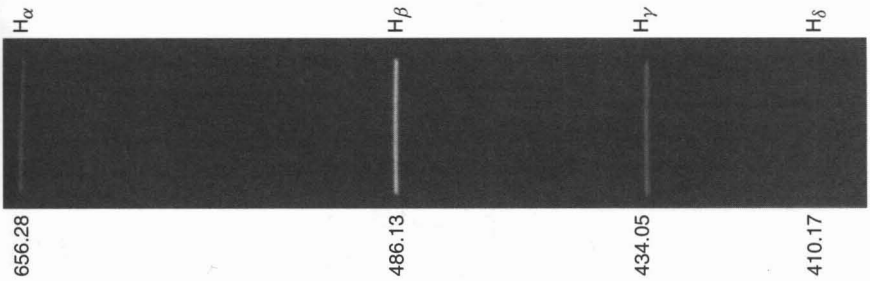


FIGURE 1.20 A spectrogram of the first four lines of the Balmer series of hydrogen as obtained with the constant-deviation spectrograph.

Figure 1.20 shows the first four lines of the Balmer series of hydrogen obtained with the "constant-deviation" spectrograph. A composite exposure containing hydrogen, sodium, and mercury lines is shown in Fig. 1.25.

1.6. THE SPECTRA OF SODIUM AND MERCURY

1.6.1. General

Mention has been made in the previous section of the spectrum of sodium (Na) and mercury (Hg); a brief analysis will be given here, since both

elements have been investigated in detail and are representative of the one-electron spectrum (Na) and two-electron spectrum (Hg) correspondingly. Sodium has 11 electrons, so that the $n = 1$ and $n = 2$ shells are completely filled and one electron ($n = 3$) is found outside closed shells. In this respect the sodium spectrum should be equivalent to that of hydrogen except for the central charge that the free electron sees. Indeed, since the nucleus with $Z = 11$ is "screened" by 10 negative charges (the $n = 1$ and $n = 2$ electrons) the free electron sees a potential $-e/r$ when far from the nucleus and a potential $(-Ze)/r + C$ when close to it, where C is the potential generated at the nucleus by the other electrons. However, whereas in hydrogen only one energy level was found for each value of n, a more complex situation arises in sodium, with several levels corresponding to the same n. This splitting is to be attributed to the fact that the time-independent Schrödinger equation for the hydrogen-like atom,

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E - V)\psi = 0,$$

admits solutions with a principal quantum number n, and angular momentum quantum number l, such that $n \geq l + 1$; when the potential that the electron sees is exactly of the Coulomb type as in the case of hydrogen, where $V = (-Ze^2)/r$ the energy eigenvalues

$$E_n = -\left[\frac{mZ^2e^4}{2(4\pi\varepsilon_0)^2\hbar^2}\right]\frac{1}{n^2} \tag{1.22}$$

are independent[15] of l, and agree with the Bohr theory. However, the screened potential that the free electron sees is no longer of the simple Coulomb type, and the energy of the level depends on both n and l. Orbits with smaller values of l are expected to come closer to the nucleus and thus be bound with greater strength; as a consequence their energy will be lower (more negative).

The energy level diagram of sodium is shown in Fig. 1.21, where the levels have been grouped according to their l value. The customary notation is used, namely, $l = 0 \rightarrow S$ state, $l = 1 \rightarrow P$ state, $l = 2 \rightarrow D$ state, $l = 3 \rightarrow F$ state, and so on, alphabetically. The last column in Fig. 1.21 gives the position of the levels of a hydrogen-like atom.

[15]This is the so-called Coulomb degeneracy: a peculiar coincidence for the Coulomb potential when used in the Schrödinger equation.

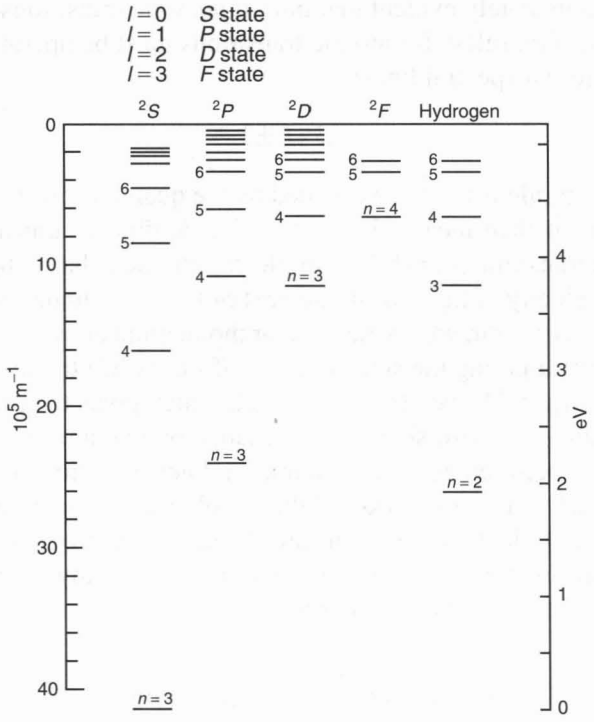


FIGURE 1.21 The energy-level diagram of sodium, grouped according to the orbital angular momentum. The last column gives the corresponding position of the levels of hydrogen. The left-hand scale is in 10^5 m^{-1}, referred to 0 for the singly ionized sodium atom; the right-hand scale is in electron volts referred to 0 at the ground state of the sodium atom.

We note that the higher the value of l, the smaller the departures from the hydrogen-like levels (as suggested qualitatively previously), and that for *given* l the energy levels for different n's follow the same ordering as the hydrogen-like atom, but with an effective charge Z^*, which for sodium is as follows: S states $Z^* \sim 11/9.6$; P states $Z^* \sim 11/10.1$; D states $Z^* \sim 1$; F states $Z^* \sim 1$.

1.6.2. Selection Rules

The spectral lines that we observe are due to transitions from one energy state to a lower one; however, in analyzing the spectrum of sodium, it

becomes immediately evident that not *all possible* transitions occur. Thus certain "selection rules" for atomic transitions must be operative, and it is found that for all spectral lines[16]

$$\Delta l = \pm 1. \tag{1.23}$$

This selection rule is readily explained by the quantum-mechanical theory of radiation; it then means that only "electric dipole" transitions occur. Indeed, the transition probability for electric dipole is larger by a factor of $(c/v)^2$ (c, velocity of light) from the next order, while under no conditions do transitions occur in which the angular momentum does not change at all ($\Delta l = 0$). By applying the selection rule of Eq. (1.23) to the energy-level diagram of Fig. 1.21, we obtain Fig. 1.22, which gives the principal lines of the sodium spectrum; since l must change by one unit, transitions will always occur between adjacent columns and never within the same one.

Figure 1.23 is a reproduction of the visible part of the above spectrum obtained by a student with the constant-deviation spectrograph. Beginning from the top (long wavelengths) we recognize the following lines (where the wavelength is given in nanometers)

(a)	Red	615.43–616.07 nm	
(b)	Yellow	589.00–589.59	(famous Na *D* lines)
(c)	Green	568.27–568.82	
(d)		514.91–515.36	
(e)		497.86–498.29	
(f)	Blue	474.80–475.19	
(g)		466.49–466.86	
(h)	Blue–Violet	449.43–449.77	

1.6.3. Fine Structure

The data in Table 1.4 on the red, yellow, and green lines of sodium, viewed with the grating, were obtained by a student simultaneously with the data used for the determination of the grating spacing d of Eq. (1.19). In the above data two wavelengths were given for each sodium line. Indeed, by viewing through the constant deviation or the grating spectrometer it is easy to resolve into a doublet each of the lines that appear in Fig. 1.23; the spacing is on the order of several tenths of a nanometer.

[16] Exceptions (such as quadrupole transitions) are found in steller spectra.

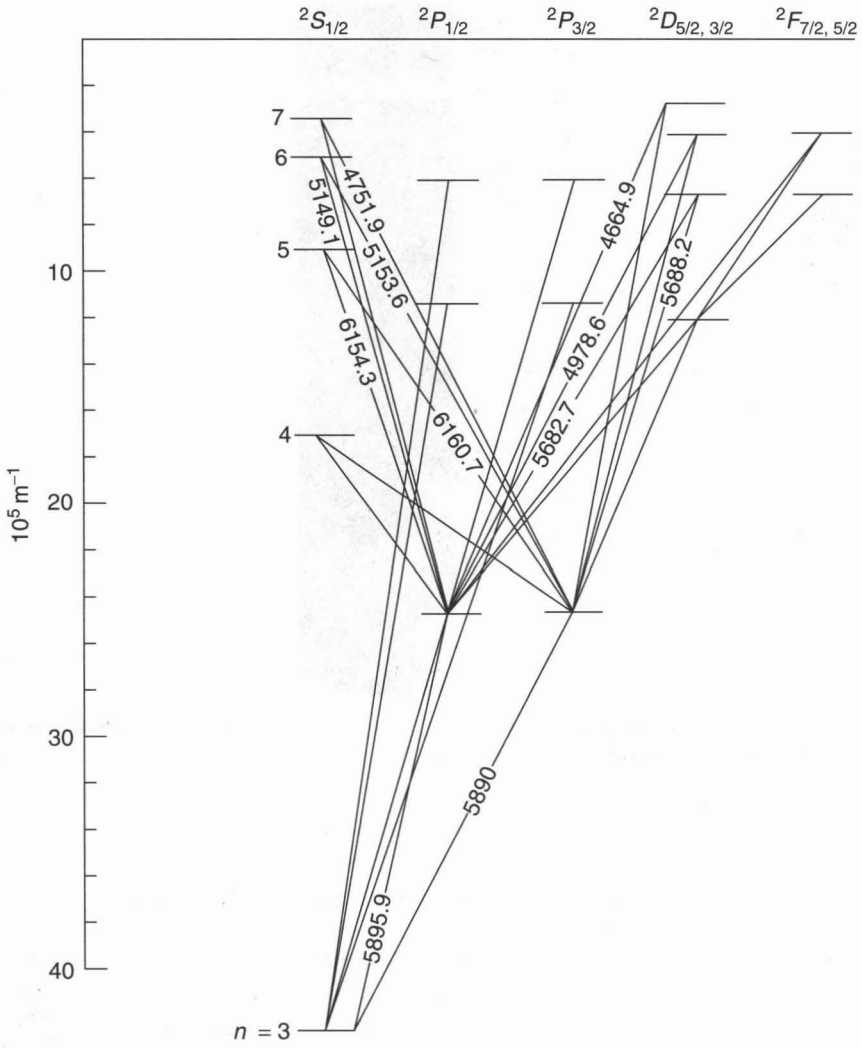


FIGURE 1.22 The "allowed" transitions between the energy levels of sodium. The wavelengths in angstroms (10 Å = 1 nm) of some of the principal lines are indicated. Note that the P states have now been shown in two columns, one referred to as $P_{1/2}$ the other as $P_{3/2}$; the small difference between their energy levels is the "fine structure."

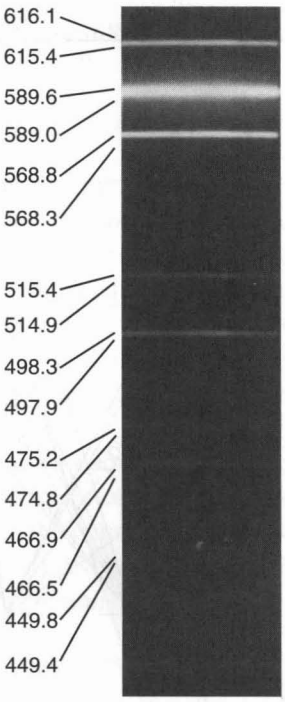


FIGURE 1.23 Photograph of the visible spectrum (in nm) of sodium as obtained with a constant-deviation spectrograph.

TABLE 1.4 Data on the Fine Structure of Sodium as Obtained with a Grating Spectrometer

Line	Order	θ_1	θ_2	$\Delta\theta$ (radians)
Red	2	41°27′	41°29′	5.8×10^{-4}
	3	55°58′	56°00′	5.8
Yellow	2	40°21′	40°23′	5.8
	3	53°49′	53°52′	8.7
	4	75°15′	75°23′	23.2
Green	2	39°32′	39°33′	2.9
	4	70°48′	70°56′	23.2

To reduce the data we note that

$$n_k\lambda = d(\sin\theta_k - \sin\theta_i),$$

where θ_i is the angle of incidence. Also

$$\theta_2 = \theta_1 + \Delta\theta$$

By letting $\sin\Delta\theta_k \approx \Delta\theta_k$, $\cos\Delta\theta_k \approx 1$,

$$n_k\Delta\lambda = d\cos\theta_k\Delta\theta_k. \tag{1.24}$$

Using $d = 3{,}692.1$ nm and averaging over orders within each line, namely writing

$$\Delta\lambda = d\frac{\sum\cos\theta_k\Delta\theta_k}{\sum n_k}, \tag{1.25}$$

we obtain for $\Delta\lambda$:

Line	$\Delta\lambda$ *(Experiment,* nm)	$\Delta\lambda$ *(Exact value,* nm)
Red	0.57	0.651
Yellow	0.63	0.597
Green	0.59	0.555

The experimental data are thus in ~10% agreement with the exact values.

This splitting of spectral lines was named "fine structure" and must reflect a splitting of the energy levels of sodium; if we express the wavelengths of the sodium lines in wave numbers ($\bar{\nu} = 1/\lambda = \nu/c$, i.e., in a scale proportional to energy since $\Delta E = hc\Delta\bar{\nu}$), it becomes evident that the spacing in all doublets is exactly the same and equal to $\Delta\bar{\nu} = 1.73 \times 10^3\ \mathrm{m}^{-1}$. Indeed, the doublet structure of all the above lines is due to the splitting of only the $3P$ ($n = 3$, $l = 1$) level as can be seen by referring back to Fig. 1.22. The splitting of the $3P$ level is due to the effect of the electron "spin" and its coupling to the orbital angular momentum (designated by l). According to the Dirac theory, the electron possesses an additional degree of freedom, called "spin," which has the properties of angular momentum of magnitude $s = \hbar/2$ (and therefore two possible orientations with respect to any axis, $m_s = +\frac{1}{2}$ or $m_s = -\frac{1}{2}$). The spin **s** can then be coupled to **l** according to the quantum-mechanical rules of addition for angular momenta; this will result in a total angular momentum of magnitude $j = l + \frac{1}{2}$ or $j = l - \frac{1}{2}$, and the energy of the state will depend on j. In the case of sodium, the $3P$ level splits into two levels, with $j = \frac{1}{2}$ and $j = \frac{3}{2}$ designated as $3P_{1/2}$ and $3P_{3/2}$ separated by $\Delta\bar{\nu} = 1.73\times 10^3\ \mathrm{m}^{-1}$.

1.6.4. Electron–Electron Coupling; the Mercury Spectrum

The mercury atom ($Z = 80$) has 80 electrons. These fill the shells $n = 1$, $n = 2$, $n = 3$, and $n = 4$ completely (60 electrons), and in addition, from the $n = 5$ shell, the $l = 0, 1, 2$ subshells account for another 18 electrons. The remaining two electrons instead of occupying the $l = 3$ and $l = 4$ subshells are in the $n = 6$ shell with $l = 0$, giving rise to a configuration equivalent to that of the helium atom.

We thus have an atom with two electrons outside closed shells in contrast to the one-electron systems of the hydrogen and sodium type. In the two-electron system, we can hardly speak of the n number of the atom, since each electron may be in a different shell; however we can still assign a total angular momentum $\mathbf{J}$ to the system, which will be the resultant of the values of each of the two electrons, and (as we saw in the previous section) of their additional degree of freedom, their spin. The addition of these four angular momenta, $\mathbf{l}_1$, $\mathbf{l}_2$, $\mathbf{s}_1$, $\mathbf{s}_2$, to obtain the resultant $\mathbf{J}$ can be done in several ways. For the helium or mercury atom, the Russell–Saunders coupling scheme holds, in which $\mathbf{l}_1$ and $\mathbf{l}_2$ are coupled into a resultant orbital angular momentum $\mathbf{L}$ and $\mathbf{s}_1$ and $\mathbf{s}_2$ into a resultant spin $\mathbf{S}$; finally $\mathbf{L}$ and $\mathbf{S}$ are coupled[17] to give the total angular momentum of the system $\mathbf{J}$. Since $\mathbf{s}_1$ and $\mathbf{s}_2$ have necessarily magnitude $\frac{1}{2}$, the resultant $\mathbf{S}$ has magnitude $S = 0$ or $S = 1$. It is customary to call the states with $S = 0$ singlets, those with $S = 1$ triplets, since when $S = 0$ for any value of L, only a single state can result, with $J = L + S = L$; when $S = 1$, however, three states can result with $J = L + S, L, L - S$, namely $J = L + 1, L, L - 1$ (provided $L \neq 0$). In systems where energy states have total angular momentum J, the selection rules for optical transitions are different, namely

$$\begin{aligned} &\Delta L = \pm 1 \\ &\Delta J = 0, \pm 1 \qquad \text{but not} \qquad J = 0 \rightarrow J = 0, \end{aligned} \tag{1.26}$$

and in principle no transitions between triplet and singlet states occur.

[17] In the ensuing discussion the quantum-mechanical rules of addition of angular momentum are used. Even if the reader is not familiar with them, he can infer them from following the development of the argument.

With these remarks in mind we consider the energy-level diagram of mercury. Since there are two electrons outside a closed shell, in the ground state they will both be in the $n = 6, l = 0$ orbit, and hence (due to the Pauli principle) must have opposite orientations of their spin, leading to $S = 0$; the spectroscopic notation is ${}^1S_0$. For the excited states one should expect both a family of singlet states and a family of triplet states; the singlets, $S = 0$, will be

$$\begin{aligned} &{}^1S_0 \quad \text{for } L = 0, \text{ and necessarily, } J = 0 \\ &{}^1P_1 \quad \text{for } L = 1, \text{ and necessarily, } J = 1 \\ &{}^1D_2 \quad \text{for } L = 2, \text{ and necessarily, } J = 2 \text{ etc.} \end{aligned}$$

Note the spectroscopic notation, where the upper left index is $2S + 1$, indicating the total spin of the state; the capital letter indicates the total L of the atom (according to the convention); and the lower right index stands for J. For the triplets, $S = 1$, and the states are

$$\begin{aligned} &{}^3S_0 \quad \text{for } L = 0, J = 1 \\ &{}^3P_{0,1,2} \quad \text{for } L = 1, J = 0, 1, 2 \\ &{}^3D_{1,2,3} \quad \text{for } L = 2, J = 1, 2, 3 \text{ etc.} \end{aligned}$$

The energy levels for mercury are shown in Fig. 1.24 with some of the strongest lines of the spectrum. It is seen that the selection rules on ΔL and ΔJ always hold, but that transitions with $\Delta S \neq 0$ do occur. It is also to be noted that the fine structure, that is, the splitting of the $6s6p\ {}^3P$ level, is of considerable magnitude: $\Delta\bar{\nu}({}^3P_0 - {}^3P_1) = 1.9 \times 10^4\ \text{m}^{-1}$; $\Delta\bar{\nu}({}^3P_1 - {}^3P_2) = 4.6 \times 10^4\ \text{m}^{-1}$. Figure 1.25 is a reproduction of the superimposed spectra of hydrogen (longest lines), mercury (medium length), and sodium (shortest lines) obtained by a student with the prism spectrograph. Beginning with long wavelengths (from the left) one identifies the following lines of mercury:

(a) Red	690.75 nm
(b) Yellow doublet	578.97–576.96
(c) Green	546.07
(d) Blue triplet	435.84
(e) Violet	404.66.

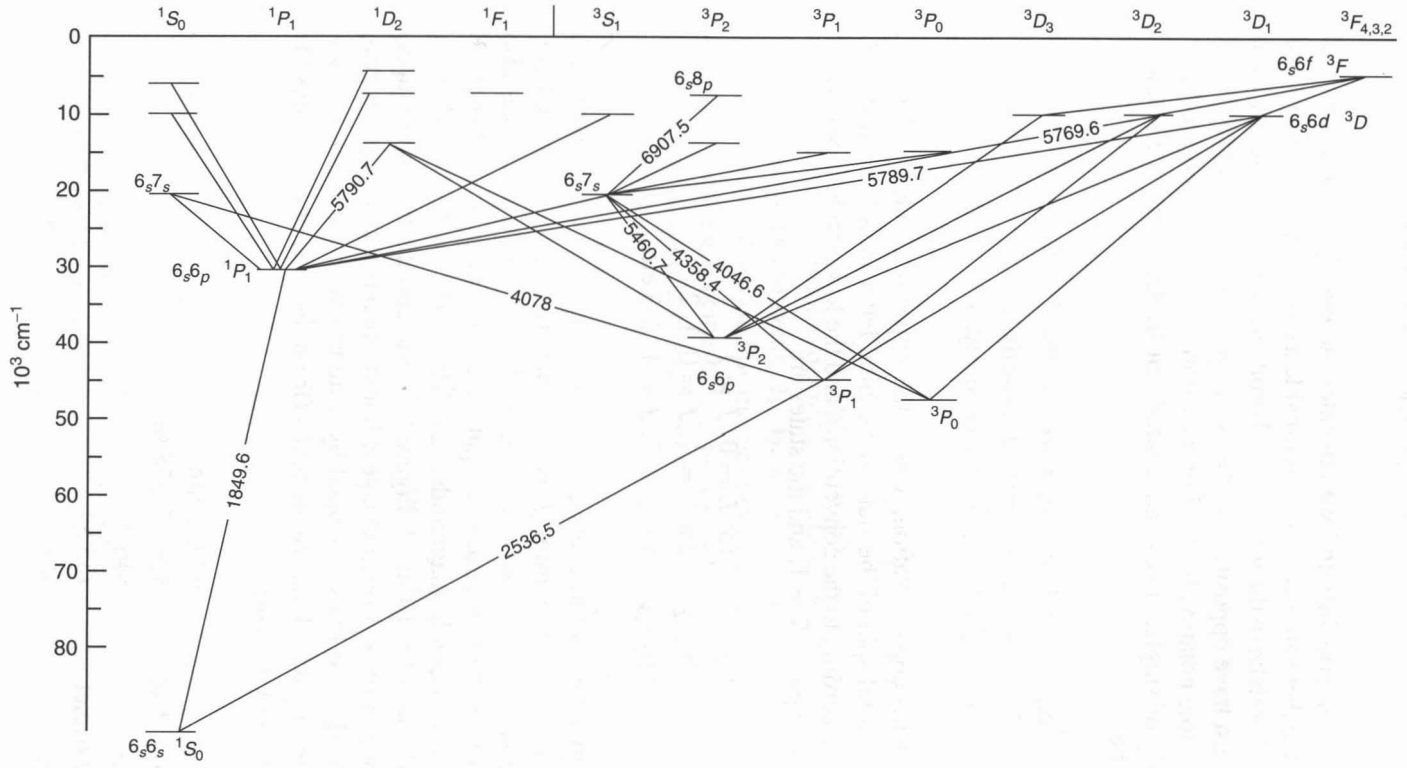


FIGURE 1.24 Energy-level diagram and the principal lines (in Å) in the spectrum of the mercury atom.

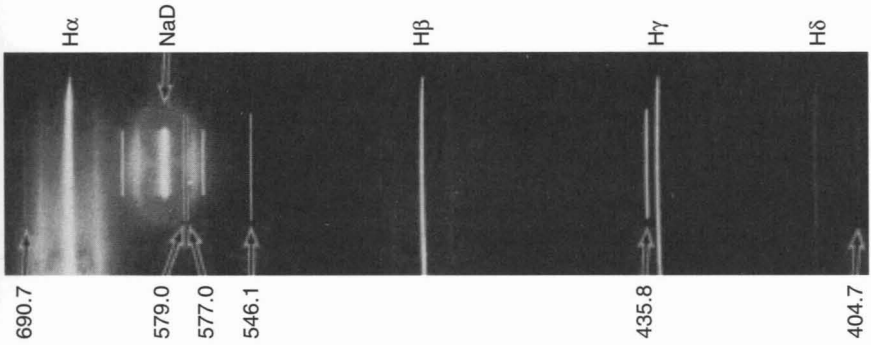


FIGURE 1.25 Photograph of the superimposed spectra of hydrogen (long slit), mercury (medium slit length), and sodium (short slit).

This concludes our discussion of the spectra and energy levels of the sodium and mercury atoms. The same treatment applies to all other one- or two-electron atoms, as well as to those with a one- or two-electron deficiency (hole) from a closed shell. Atoms with more electrons outside closed shells are treated on analogous lines, but the coupling schemes become more complicated, giving rise, as in the case of the rare earths, to extremely complex spectra.